**NANYANG TECHNOLOGICAL UNIVERSITY**

**SEMESTER I EXAMINATION 2017-2018**

**MH2500 – Probability and Introduction to Statistics**

December 2017 TIME ALLOWED: 2 HOURS

---

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages including appendix.
2. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
3. Answer each question beginning on a **FRESH** page of the answer book.
4. This is a **RESTRICTED OPEN BOOK** exam. Candidates are allowed **TWO** double-sided A4 size help sheets.
5. Candidates may use calculators. However, they should write down systematically the steps in the workings.
6. The table with values of the standard normal distribution is included at the end of this examination paper.

## QUESTION 1 **(20 marks)**

Suppose that $X$ is a random variable taking values in $\{0, 1, 2\}$ and $Y$ is a random variable taking values in $\{0, 1, 2, 3\}$. A partial joint probability table of $X$ and $Y$ is given below.

| | $Y = 0$ | $Y = 1$ | $Y = 2$ | $Y = 3$ |
|---|---|---|---|---|
| $X = 0$ | 0.10 | | 0.16 | |
| $X = 1$ | 0.05 | 0.04 | 0.08 | 0.04 |
| $X = 2$ | 0.10 | 0.08 | | 0.08 |

(a) (5 marks) Find the probability $\Pr(X > Y)$.

(b) (5 marks) Find the conditional mass function of $X$ given that $Y = 0$.

(c) (5 marks) If the marginal distribution of $Y$ is uniform, what is the probability $\Pr(X = 0, Y = 1)$?

(d) (5 marks) Find the probability $\Pr(X = 1)$ and determine whether $X$ and $Y$ are independent when the marginal distribution of $Y$ is uniform.

[Solution:]

(a) $\Pr(X > Y) = \Pr(X = 1, Y = 0) + \Pr(X = 2, Y = 0) + \Pr(X = 2, Y = 1) = 0.05 + 0.10 + 0.08 = 0.23$.

(b) The conditional mass function is

$$p_{X|Y=0}(x) = \begin{cases} 0.4, & x = 0; \\ 0.2, & x = 1; \\ 0.4, & x = 2. \end{cases}$$

(c) The column for $Y = 1$ would sum up to 0.25, giving that $\Pr(X = 0, Y = 1) = 0.13$.

(d) $\Pr(X = 1) = 0.21$ by summing the entries in the row corresponding to $X = 1$, and since $\Pr(Y = 1) = 0.25$, we know that $\Pr(X = 1, Y = 0) \neq \Pr(X = 1)\Pr(Y = 0)$, so $X$ and $Y$ are not independent.

**QUESTION 2** **(15 marks)**

Suppose you have in your pocket a fair coin and a two-headed coin. You select one of the coins at random (with equal probability). All questions below are about this one coin; it is not replaced, and no other coin is drawn.

(a) (5 marks) When you flip it, it shows head. What is the probability that it is the fair coin?

(b) (5 marks) What is the probability that it will show a head on a second flip, given that it shows a head on the first flip?

(c) (5 marks) When you flip the coin a second time, it shows head again. What is the probability that it is the fair coin?

Leave your answers as fractions.

[Solution:]

(a) By Bayes' Theorem,

$$\begin{aligned}\Pr(\text{fair}|H) &= \frac{\Pr(H|\text{fair})\Pr(\text{fair})}{\Pr(H|\text{fair})\Pr(\text{fair}) + \Pr(H|\text{two-headed})\Pr(\text{two-headed})} \\ &= \frac{\frac{1}{2}\cdot\frac{1}{2}}{\frac{1}{2}\cdot\frac{1}{2} + 1\cdot\frac{1}{2}} \\ &= \frac{1}{3}.\end{aligned}$$

(b) Now we update our prior probability to $\Pr(\text{fair}) = 1/3$ and $\Pr(\text{two-headed}) = 2/3$. Therefore the desired probability is

$$\frac{1}{2}\Pr(\text{fair}) + 1\cdot\Pr(\text{two-headed}) = \frac{5}{6}.$$

(c) Similarly to (a), by Bayes' Theorem again with our new priors,

$$\begin{aligned}\Pr(\text{fair}|H) &= \frac{\Pr(H|\text{fair})\Pr(\text{fair})}{\Pr(H|\text{fair})\Pr(\text{fair}) + \Pr(H|\text{two-headed})\Pr(\text{two-headed})} \\ &= \frac{\frac{1}{2}\cdot\frac{1}{3}}{\frac{1}{2}\cdot\frac{1}{3} + 1\cdot\frac{2}{3}} \\ &= \frac{1}{5}.\end{aligned}$$

[Alternative Solution:] We can also compute the probabilities based on the original prior probabilities for parts (b) and (c).

(b) By the law of total probability,

$$\begin{aligned}\Pr(\text{two flips are heads}) &= \Pr(\text{two flips are heads}|\text{fair})\Pr(\text{fair}) \\ &\quad + \Pr(\text{two flips are heads}|\text{two-headed})\Pr(\text{two-headed}) \\ &= \frac{1}{4}\cdot\frac{1}{2} + 1\cdot\frac{1}{2} \\ &= \frac{5}{8}\end{aligned}$$

and

$$\begin{aligned}\Pr(\text{first flip is head}) &= \Pr(\text{first flip is head}|\text{fair})\Pr(\text{fair}) \\ &\quad + \Pr(\text{first flip is head}|\text{two-headed})\Pr(\text{two-headed}) \\ &= \frac{1}{2}\cdot\frac{1}{2} + 1\cdot\frac{1}{2} \\ &= \frac{3}{4}\end{aligned}$$

Finally,

$$\Pr(\text{second flip is head}|\text{first flip is head}) = \frac{\Pr(\text{two flips are heads})}{\Pr(\text{first flip is head})} = \frac{\frac{5}{8}}{\frac{3}{4}} = \frac{5}{6}.$$

(c) By Bayes' Theorem,

$$\begin{aligned}\Pr(\text{fair}|HH) &= \frac{\Pr(HH|\text{fair})\Pr(\text{fair})}{\Pr(HH|\text{fair})\Pr(\text{fair}) + \Pr(HH|\text{two-headed})\Pr(\text{two-headed})} \\ &= \frac{\frac{1}{4}\cdot\frac{1}{2}}{\frac{1}{4}\cdot\frac{1}{2} + 1\cdot\frac{1}{2}} \\ &= \frac{1}{5}.\end{aligned}$$

**QUESTION 3** **(20 marks)**

A random rectangle is formed in the following way: The base $X$ is chosen to be uniformly random on $[0,1]$, and after having generated the base, the height $H$ is chosen to be uniform on $[0,X]$.

(a) (5 marks) Find the conditional expectation $\mathbb{E}(H|X)$.

(b) (5 marks) Find the expected area of the rectangle.

(c) (5 marks) Find the variance of the area of the rectangle.

(d) (5 marks) Find the correlation between the base and the area of the rectangle.

Leave your answers as fractions, or numerical values that are accurate up to 0.001.

[Solution:] Let $H$ denote the height of the rectangle.

(a) $\mathbb{E}(H|X) = X/2$.

(b) The area is $XH$ and thus

$$\mathbb{E}(XH) = \mathbb{E}(\mathbb{E}(XH|X)) = \mathbb{E}(X\,\mathbb{E}(H|X)) = \mathbb{E}\,\frac{X^2}{2} = \frac{1}{2}\int_0^1 x^2 dx = \frac{1}{6}.$$

(c) Similarly to part (b),

$$\mathbb{E}[(XH)^2] = \mathbb{E}(X^2\,\mathbb{E}(H^2|X)) = \mathbb{E}\left(X^2 \cdot \frac{1}{3}X^2\right) = \frac{1}{3}\int_0^1 x^4 dx = \frac{1}{15}\ (\approx 0.0667)$$

and thus

$$\mathrm{Var}(XH) = \mathbb{E}[(XH)^2] - (\mathbb{E}(XH))^2 = \frac{7}{180}\ (\approx 0.0389).$$

(d) We know that $\mathbb{E}(X) = 1/2$ and $\mathrm{Var}(X) = 1/12$. We compute that

$$\begin{aligned}
\mathrm{cov}(X, XH) &= \mathbb{E}(X^2 H) - \mathbb{E}(X)\,\mathbb{E}(XH) \\
&= \mathbb{E}(X^2\,\mathbb{E}(H|X)) - \frac{1}{2}\cdot\frac{1}{6} \\
&= \frac{1}{2}\,\mathbb{E}\,X^3 - \frac{1}{12} \\
&= \frac{1}{2}\cdot\frac{1}{4} - \frac{1}{12} \\
&= \frac{1}{24}.
\end{aligned}$$

Therefore the correlation

$$\rho(X, XH) = \frac{\operatorname{cov}(X, XH)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(XH)}} = \frac{\frac{1}{24}}{\sqrt{\frac{1}{12} \cdot \frac{7}{180}}} = \frac{1}{2}\sqrt{\frac{15}{7}} \ (\approx 0.7319).$$

**QUESTION 4** **(25 marks)**

Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables of the following density function:

$$f(x) = \begin{cases} Cx^{-(1+\alpha)}, & \text{if } x \geq 1; \\ 0, & \text{otherwise,} \end{cases}$$

where $\alpha > 0$ is a constant and $C$ is the normalization constant. Let $Z_n$ be the geometric mean of $X_1, \ldots, X_n$, that is, $Z_n = (X_1 X_2 \cdots X_n)^{1/n}$.

(a) (5 marks) Determine the value of $C$. (The value of $C$ may depend on $\alpha$, in which case your answer should be expressed in terms of $\alpha$).

(b) (10 marks) Find the density function of $Y_i = \ln X_i$.

(c) (5 marks) Determine whether $Z_n$ converges to some constant $L$ in probability as $n \to \infty$. If yes, find the constant $L$. (Hint: you may find part (b) useful).

(d) (5 marks) If your answer to part (c) is 'yes', show that

$$\Pr\left(Z_n \leq e^{\frac{1}{\alpha}} L\right) \geq 1 - \frac{1}{n}.$$

If your answer to part (c) is 'no', show that

$$\Pr(Z_n \geq e^{\frac{1}{\alpha}} n) \geq 1 - \frac{1}{n}.$$

[Solution:]

(a) Since

$$\int_1^\infty \frac{dx}{x^{\alpha+1}} = \frac{1}{\alpha},$$

it must hold that $C = \alpha$.

(b) We first find the cumulative function of $Y_i$. Note that $Y_i \geq 0$.

$$\Pr(Y_i \leq y) = \Pr(X_i \leq e^y) = \alpha \int_1^{e^y} \frac{dx}{x^{\alpha+1}} = 1 - e^{-\alpha y}, \quad y \geq 0.$$

Differentiating yields the desired density function,

$$p(y) = \alpha e^{-\alpha y}, \quad y \geq 0.$$

(c) Observe that

$$\ln Z_n = \frac{Y_1 + \cdots + Y_n}{n}.$$

Note that $Y_i$ is an exponential variable with parameter $\alpha$, both $\mathbb{E}(Y_i) = 1/\alpha$ and $\operatorname{Var}(Y_i) = 1/\alpha^2$ exist, it follows from the weak law of large numbers that $\ln Z_n \to 1/\alpha$ in probability. Therefore $Z_n$ converges in probability to

$$L = e^{1/\alpha}.$$

(d) Note that

$$\operatorname{Var}(\ln Z_n) = \frac{\operatorname{Var}(Y_i)}{n} = \frac{1}{n\alpha^2}.$$

By Chebyshev's inequality,

$$\Pr(e^{-t}L \leq Z_n \leq e^t L) = \Pr(|\ln Z_n - \ln L| \leq t) \geq 1 - \frac{\operatorname{Var}(\ln Z_n)}{t^2} = 1 - \frac{1}{n\alpha^2 t^2}.$$

Letting $t = 1/\alpha$ yields the desired result.

**QUESTION 5** **(20 marks)**

A worker in a large supermarket chain was tasked to find out whether a particular brand of milk sold in 1000 millilitres packs contains less milk than claimed. He took a random sample of 144 packs of milk and found that the sample mean is 990 millilitres of milk. Assume that the population standard deviation is 48 millilitres.

(a) (10 marks) Use the 0.05 level of significance to test if the milk content is less than 1000 millilitres.

(b) (10 marks) If the true population mean is 995.5 millilitres, compute the type II error $\beta$ for the test in Part (a). Leave your answer as a numerical value that is accurate up to 0.001.

[Solution:]

(a) $H_0 : \mu = 1000$

$H_1 : \mu < 1000.$

Test statistic $z$:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{990 - 1000}{48/\sqrt{144}} = -2.5$$

Using significance level $\alpha = 0.05$, the rejection region will be when $z < -1.645$.

Conclusion: Since $z = -2.5 < -1.645$, we reject $H_0$. Therefore based on the evidence, the average volume of milk in the 1000 millilitres packs is less than 1000 millilitres.

(b) The rejection region is the same, $z < -1.645$. The values of $\bar{x}$ for which we reject $H_0$ is

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < -1.645,$$

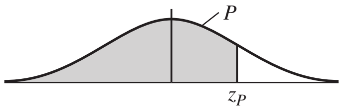
i.e.

$$\frac{\bar{x} - 1000}{48/\sqrt{144}} < -1.645,$$

i.e. $\bar{x} < 993.42$.

$$\begin{aligned}
\Pr(\text{committing a type II error}) &= \Pr(\text{ do not reject } H_0 \text{ when } H_0 \text{ is not true })\\
&= \Pr(\bar{x} > 993.42 \text{ when } \mu = 995.5)\\
&= \Pr\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} > \frac{993.42 - 995.5}{48/\sqrt{144}}\right)\\
&= \Pr(z > -0.52)\\
&= 0.6985.
\end{aligned}$$

Cumulative Normal Distribution — Values of $P$ corresponding to $z_p$ for the Standard Normal Curve.




**$z$ is the standard normal variable. The value of $P$ for $-z_p$ equals 1 minus the value of $P$ for $+z_p$; for example, the $P$ for –1.62 equals 1 – .9474 = .0526.**

| $z_p$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| .0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| .1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| .2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| .3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| .4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| .5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| .6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| .7 | .7580 | .7611 | .7642 | .7673 | .7704 | .7734 | .7764 | .7794 | .7823 | .7852 |
| .8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 | .8106 | .8133 |
| .9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9761 | .9767 |
| 2.0 | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9979 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9982 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |
| 3.0 | .9987 | .9987 | .9987 | .9988 | .9988 | .9989 | .9989 | .9989 | .9990 | .9990 |
| 3.1 | .9990 | .9991 | .9991 | .9991 | .9992 | .9992 | .9992 | .9992 | .9993 | .9993 |
| 3.2 | .9993 | .9993 | .9994 | .9994 | .9994 | .9994 | .9994 | .9995 | .9995 | .9995 |
| 3.3 | .9995 | .9995 | .9995 | .9996 | .9996 | .9996 | .9996 | .9996 | .9996 | .9997 |
| 3.4 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9998 |

**END OF PAPER**